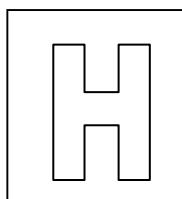


Candidate Name: _____

Class Adm No

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2018 Preliminary Exams Pre-University 3

BIOLOGY

9744/02

Paper 2 Structured Questions

14 September 2018

2 hours

READ THESE INSTRUCTIONS FIRST

Do not open this booklet until you are told to do so.

Write your Admission number and name on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions in the question booklet.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question. At the end of the examination, fasten all your work securely together.

For Examiner's Use	
1	
2	
3	
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7	
8	
Total	

This question paper consists of 28 printed pages, including 3 blank pages.

[Turn over

Answer **all** questions in this section.

1. The survival and growth of living things are largely affected by the biotic and abiotic factors of the ecosystem. To investigate how temperature of the habitat affects growth of an animal, the relationship between mean mass of different mammalian species and the mean annual temperature of their habitat were investigated. Fig. 1.1 shows the result.

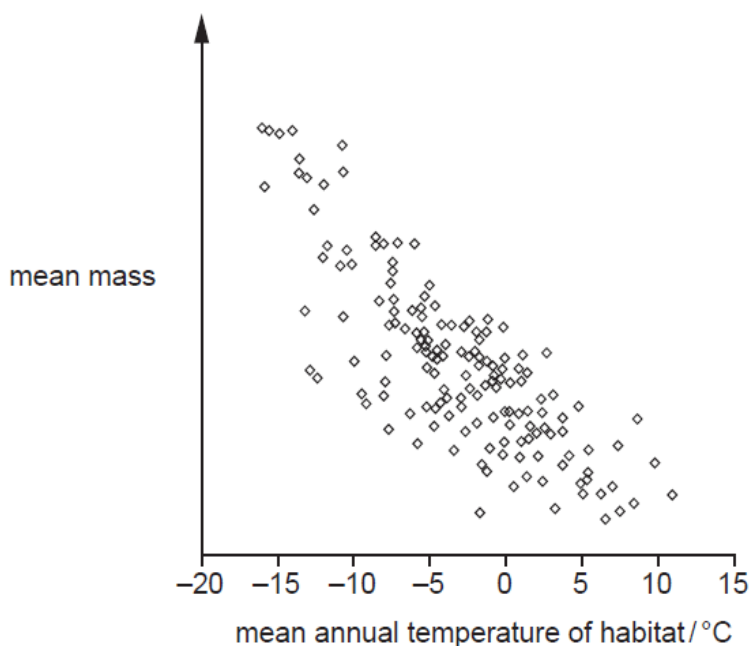


Fig. 1.1

- (a) State what each plot point in the graph represents.

.....[1]

The mass of a mammal is commonly associated with the mass of fats. The fatty layer under the skin serves many important functions.

- (b) Describe the formation of ester bond in triglyceride.

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[2]

- (c) Discuss the relationship between the mean mass of mammalian species and the mean annual temperature of their habitat.

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The mean mass of the mammal is studied instead of the individual mass of the mammal, due to the variation in mass between mammals.

- (d) Explain why this is important for survival of species.

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[Total:10]

2. Since the start of the epidemic, Human immunodeficiency virus (HIV) has caused 35 million deaths worldwide.

(a) Describe the structural components of virus.

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(b) Distinguish between the reproductive cycle of HIV and influenza virus.

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- (c)** Explain how HIV infection causes disease in humans.

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One common treatment method for HIV infected patients is the use of reverse transcriptase inhibitors. These inhibitors bind to the inhibitor binding site on the reverse transcriptase.

However, HIV often develops resistance against these inhibitors due to a single base substitution in the DNA sequence, resulting in a change of an amino acid in a subunit of reverse transcriptase. The mutant protein has the same number of amino acid sequence as the wild type protein and retains reverse transcriptase activity, but does not allow the binding of the inhibitors.

- (d)** Explain why the number of amino acid in mutated protein remains unchanged and retains reverse transcriptase activity, but does not allow the binding of the inhibitors.

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[Total:13]

3. Fig. 3.1 shows a plant cell undergoing gametogenesis.

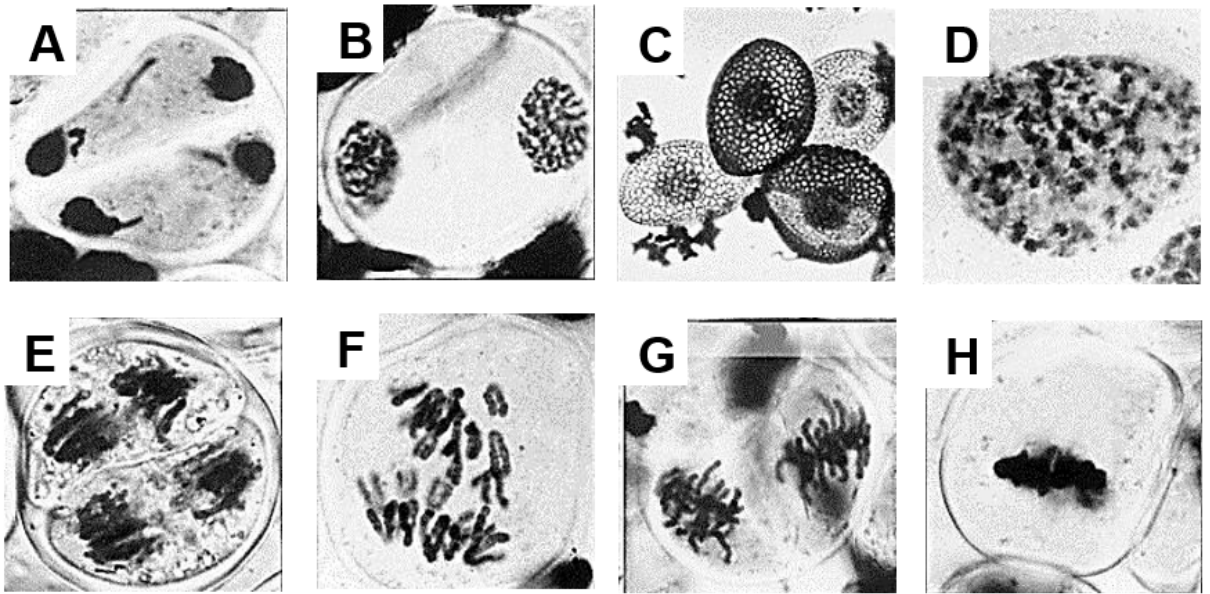


Fig. 3.1

- (a) Arrange the letters **A** to **H** to show the correct sequence of events for this nuclear division.

.....[2]

The cell was exposed to a toxin that causes spindle fibres to increase in length.

- (b)** Explain how the cell will be affected if the toxin was administered towards the end of stage **H**.

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- (c)** Explain how the toxin may lead to an absence of daughter cells or daughter cells with varied genetic content.

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- (d)** Explain the significance of this nuclear division during gametogenesis.

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[Total:10]

4. During an investigation, pollen grains from the anthers of a pure-bred plant with green leaves and hairy fruit were transferred by a student to the stigmas of a pure-bred plant with mottled (green and yellow) leaves and smooth-surfaced fruits. All the F_1 generation had green leaves and smooth fruit.

(a) Explain what is meant by pure-bred.

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[1]

(b) Explain how the genotype is linked to phenotype in tomato.

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[2]

A test cross was then made between the F_1 plant and a plant with mottled leaves and hairy fruit. The phenotype of 100 offsprings of the test cross were recorded and shown in Table 4.1.

Table 4.1

Green leaves, smooth fruit	Green leaves, hairy fruit	Mottled leaves, smooth fruit	Mottled leaves, hairy fruit
8	46	40	6

- (c)** Using suitable symbols, draw a genetic diagram to explain the results of the test cross.

- (d) Explain the difference between the observed test cross results from the expected test cross ratio.

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[2]

In a separate investigation, a test cross was performed between the same F1 plant and plant with mottled leaves and hairy fruit. Instead of recording the results for all four groups of phenotypes, the student recorded the number of the two more common groups of phenotype among the 100 offsprings. The results were shown in Table 4.2.

Table 4.2

Green leaves, smooth fruit	Green leaves, hairy fruit	Mottled leaves, smooth fruit	Mottled leaves, hairy fruit
<i>Not recorded</i>	52	44	<i>Not recorded</i>

The student claimed that the observed test cross ratio of offspring can be simplified to 1:1.

- (e) Suggest why the student made this claim.

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[1]

- (f) Perform the chi-squared test to verify this claim.

The formula and probability table is given as follow.

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

Degrees of freedom	Probability		
	0.10	0.05	0.01
1	2.71	3.84	6.64
2	4.69	5.99	9.21
3	6.25	7.82	11.35
4	7.78	9.49	13.28

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[Total:14]

5. Fig. 5.1 shows an electron micrograph of an organelle.

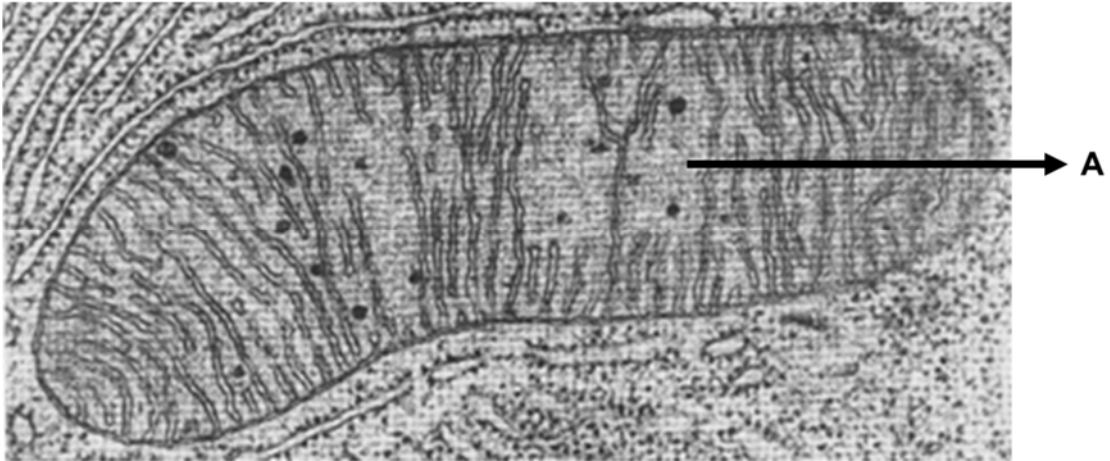


Fig. 5.1

- (a)** State the stage(s) of aerobic respiration that occur in **A**.

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- (b)** List down the end products of the stage(s) identified in **(a)**.

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[2]

- (c)** Describe the importance of membrane during aerobic respiration.

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[3]

Anaerobic respiration occurs in animals, plants, bacteria and archaea. It occurs when there is an absence of oxygen, releasing less amount of energy as compared to aerobic respiration.

(d) Explain how anaerobic respiration releases energy.

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- (e) Many enzymes are involved in the different stages of aerobic and anaerobic respiration.

Fig. 5.2 shows the changes in energy during the progress of an uncatalyzed reaction.

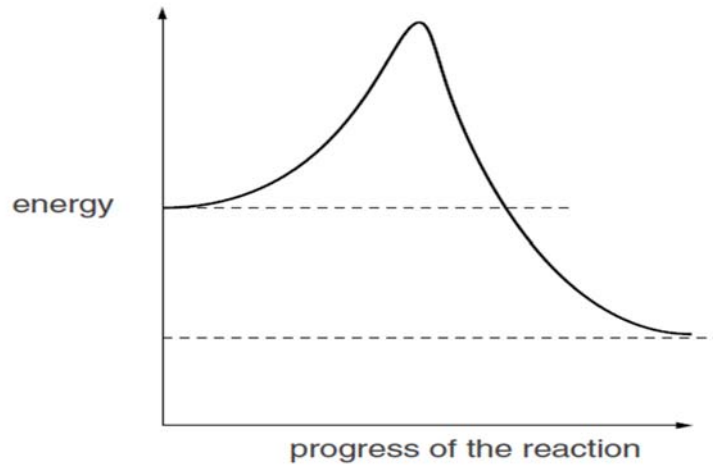


Fig. 5.2

- (i) Draw on Fig. 5.2 the changes in energy during enzyme-catalysed reaction.

[1]

- (ii) Label the following terms on your drawn diagram.

- Activation energy
- Overall energy released

[1]

[Total:12]

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6. Fig. 6.1 shows the effects of temperature and light intensity on the rate of photosynthesis of a plant.

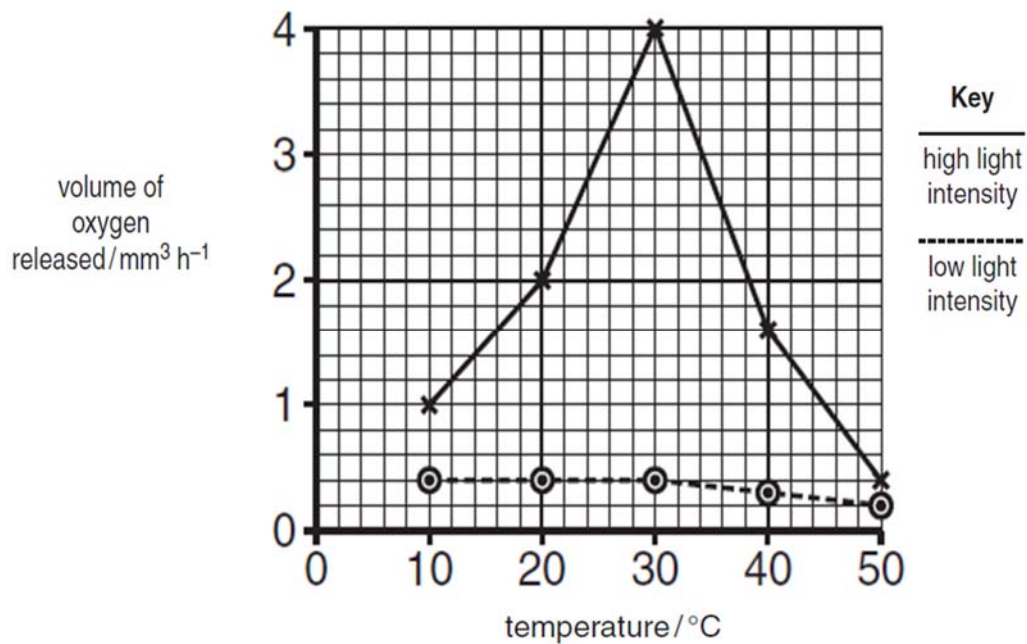


Fig. 6.1

- (a) Compare the effect of temperature on the rate of photosynthesis between low and high light intensity.

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7. Recent research has revealed connection between the size of the nucleolus and the lifespan of the cell. Researchers believed that the nucleolus is an important organelle that coordinates many processes. This may imply that it is involved in many different signalling pathways.

(a) Suggest why the size of the nucleolus varies in different tissues.

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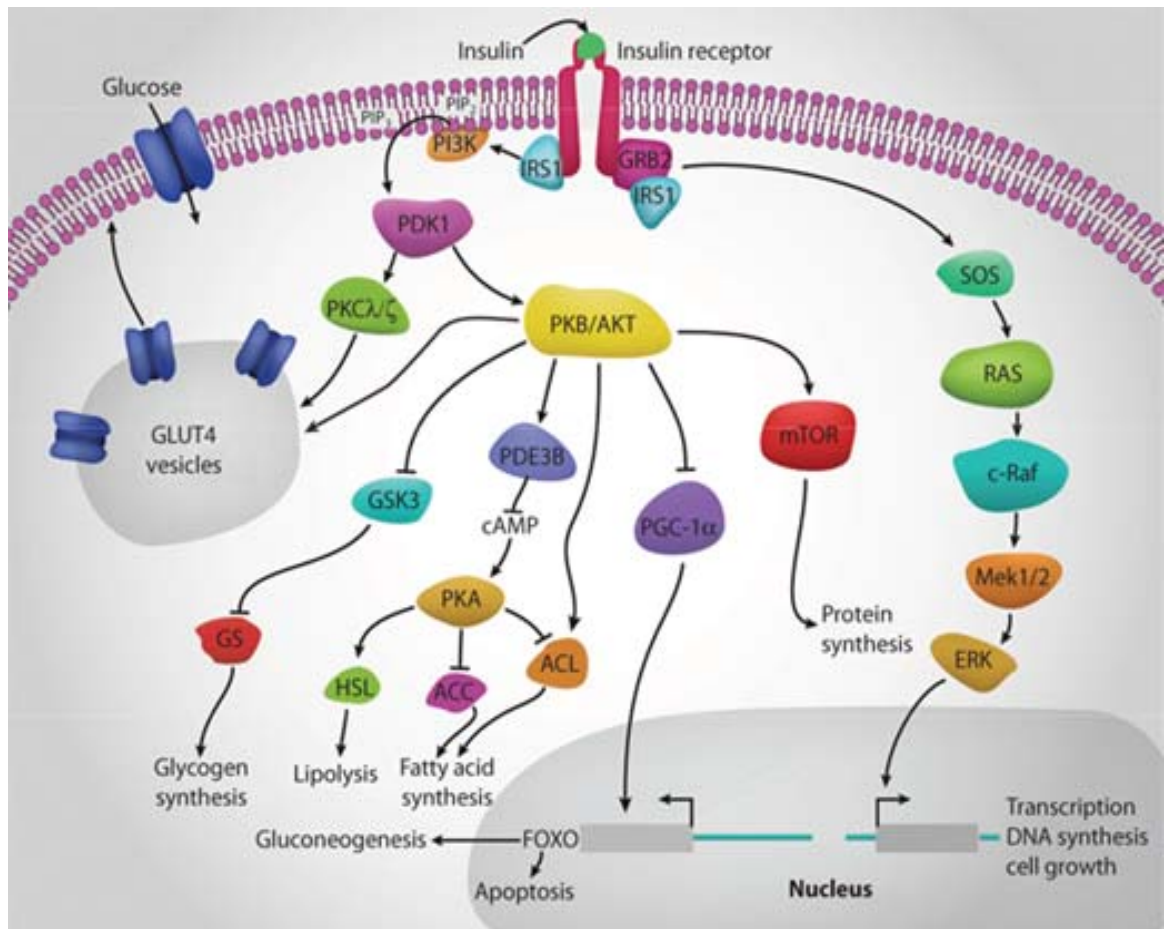
Reduced insulin signalling, dietary restriction and other signalling pathways increase the activity of a gene known as *NCL-1*, a nucleolar regulating gene. The increase in gene expression of *NCL-1* is associated with reduced size of nucleolus and increased lifespan.

(b) Describe how expression of *NCL-1* gene can be increased at the chromatin level in eukaryotes.

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Insulin has several effects on cellular growth and metabolism. The actions of insulin are mediated by the insulin receptor, and the end results vary in different cells. Fig. 7.1 shows the signalling pathways activated when insulin binds to the receptor.



Michael W.King: Integrative Medical Biochemistry

Fig. 7.1

The association between insulin signalling and cancer risk is widely debated by scientists due to the release of contradicting studies.

- (c)** With reference to Fig. 7.1, suggest how dysregulation of the insulin signalling pathway may lead to cancer.

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- (d)** Suggest why dysfunctional insulin receptors is linked to many abnormalities.

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[Total:12]

8. Tuberculosis is an infectious disease caused by the intracellular pathogen, *Mycobacterium tuberculosis*. The disease affects the respiratory tract and caused an estimated 1.6 million deaths in 2016. Tuberculosis is treated with antibiotics such as rifampicin.

Increasingly, bacteria are developing resistance towards rifampicin. This is because bacteria contain an enzyme, rifampicin monooxygenase, which changes the structure of rifampicin from cyclic to linear, and thus inactivating the antibiotics.

- (a) Define antibiotics.

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.....[2]

- (b) Suggest how scientists should design a new drug to overcome the problem with rifampicin resistance.

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.....[1]

LRRK2 is a large protein encoding seven functional domains as shown in Fig. 8.1. LRRK2 has 2527 amino acids and a molecular weight of 286 kDa. It is a multi-domain protein comprising of a catalytic core domain, kinase domain and five protein-protein interaction domains. Mutations in the gene encoding for LRRK2 are associated with Parkinson's disease, cancer, Crohn's disease and bacterial infections.

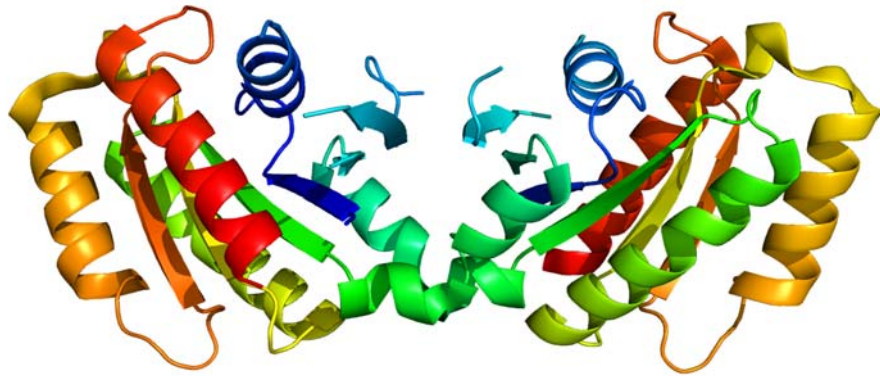


Fig. 8.1

- (c) Using the information above, explain why there are more than one disease associated with mutations in the gene coding for LRRK2.

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Studies have shown that LRRK2 inhibits the maturation of phagosome in macrophages.

- (d) Explain what will happen when an individual infected with *Mycobacterium tuberculosis* have functional LRRK2 protein.

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Fig. 8.2 shows the organisation of the DNA sequence which encode one of the domains of human LRRK2. The size of the exon and intron is stated within the brackets.

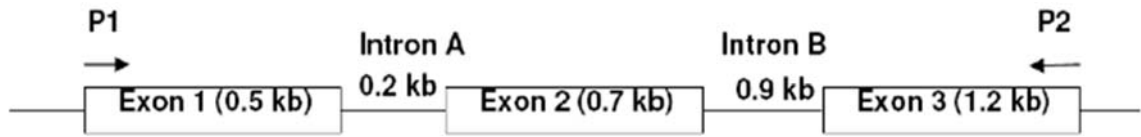


Fig. 8.2

The DNA sequences from three individuals, normal, mutant **A** and mutant **B** were obtained. The DNA sequences were subjected to transcription to form mRNA. The mRNA then underwent reverse transcription to form complementary DNA (cDNA). Polymerase chain reaction (PCR) was performed using primer **P1** and **P2**, which were specific to the exon regions shown in Fig. 8.2.

(e) Explain why cDNA sequences were used instead of DNA sequence.

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[2]

(f) Explain why two primers are needed for the polymerase chain reaction.

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[2]

The products of PCR were subjected to electrophoresis on an agarose gel and the results are shown in Fig. 8.3.

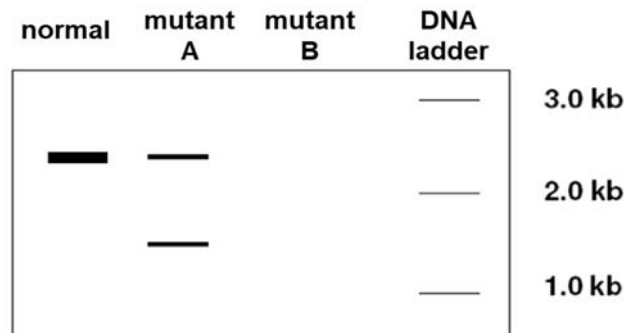


Fig. 8.3

- (g)** Deduce the size of the PCR product for the normal individual.

Show your working in the space below.

[1]

- (h)** Account for the band patterns observed for

- (i) mutant A**

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[2]

- (ii) mutant B**

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[2]

- (i) Explain why there are more than one bands in the DNA ladder.

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[Total:21]

End of Paper